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Synergistic/antagonistic effects of nitrate/ammonium enrichment on fatty acid biosynthesis and translocation in coral under heat stress

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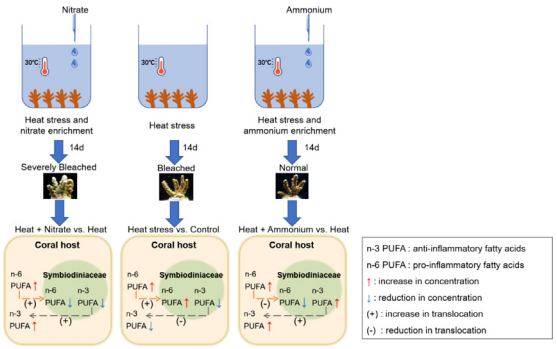
HIGHLIGHTS

- Coral's fatty acids showed opposite responses to NO_3^- and NH_4^+ under heat stress.
- NH_4^+ promotes anti-inflammatory fatty acids biosynthesis in Symbiodiniaceae.
- NO_3^- promotes pro-inflammatory FA biosynthesis in coral.
- NH_4^+ is antidotal but NO_3^- is toxic for corals under heat stress.

GRAPHICAL ABSTRACT



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ABSTRACT

Superimposed on ocean warming, nitrogen enrichment caused by human activity puts corals under even greater pressure. Biosynthesis of fatty acids (FA) is crucial for coral holobiont survival. However, the responses of FA biosynthesis pathways to nitrogen enrichment under heat stress in coral hosts and Symbiodiniaceae remain unknown, as do FA translocation mechanisms in corals. Herein, we used the thermosensitive coral species *Acropora hyacinthus* to investigate changes in FA biosynthesis pathways and polyunsaturated FA translocation of coral hosts and Symbiodiniaceae with respect to nitrate and ammonium enrichment under heat stress. Heat stress promoted pro-inflammatory FA biosynthesis in coral hosts and inhibited FA biosynthesis in Symbiodiniaceae. Nitrate enrichment inhibited anti-inflammatory FA biosynthesis in Symbiodiniaceae, and promoted pro-inflammatory FA biosynthesis in coral hosts and translocation to Symbiodiniaceae, leading to bleaching after 14 days of culture. Intriguingly, ammonium enrichment promoted anti-inflammatory FA biosynthesis in Symbiodiniaceae and translocation to hosts, allowing corals to better endure heat stress. We constructed schematic diagrams of the shift in FA biosynthesis and translocation in and between *A. hyacinthus* and its Symbiodiniaceae under heat stress, heat and nitrate co-stress, and heat and ammonium co-stress. The findings provide insight into the mechanisms of coral bleaching under environmental stress from a fatty acid perspective.

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1. Introduction

Coral reefs are oases of productivity in oligotrophic marine areas (DeCarlo et al., 2020), rich in biodiversity and economic value (Cybulski et al., 2020). Climate change is one of the most significant challenges to the survival of coral reefs. Continued increases in sea temperature can lead to bleaching and death of reef-building corals, fundamentally altering the structure and function of coral reefs (Hoey et al., 2016). Environmental pollution is another major factor contributing to the degradation of coral reefs (Wong et al., 2017). Nutrient enrichment can stress corals and affect their health (McCauley et al., 2015), reducing the area covered by living reef corals (Darling et al., 2019). Coverage of living reef corals is lowest when the dissolved inorganic nitrogen (DIN) enrichment in seawater exceeds 9 μM (Duprey et al., 2016). Coral communities in eutrophic waters are dominated by stress-tolerant corals, with competitive corals such as *Acropora* rarely present (Huang et al., 2011; Cybulski et al., 2020). Wastewater discharge and excessive nutrient inputs for agricultural production may be among the major causes of offshore eutrophication and degradation of coral reef ecosystems (Wang et al., 2020; Pan et al., 2021; Zhao et al., 2021).

Symbiodiniaceae are endosymbiotic photosynthetic algae that give photosynthetic fixed carbon to their coral host (Papina et al., 2003), and the coral host provides inorganic nutrients for Symbiodiniaceae in exchange (Rädecker et al., 2015). Heat stress can cause damage to both coral host and Symbiodiniaceae cells, resulting in reactive oxygen species (ROS) accumulation (Weis, 2008). Elevated nutrient concentrations in the reef environment increase the sensitivity of corals to heat stress (Wiedenmann et al., 2013; Vega Thurber et al., 2014; Donovan et al., 2020). Coral hosts may control the rate of growth and maintenance of high photosynthetic rates of Symbiodiniaceae through nitrogen limitation (Yellowlees et al., 2008). An increase in nutrients stimulates the growth of Symbiodiniaceae, leading to higher Symbiodiniaceae density in corals (Morris et al., 2019). However, the effects of different forms of nitrogen (nitrate or ammonium) enrichment on corals under heat stress are believed to vary. Reactive oxygen species (ROS), nitric oxide (NO), lipid peroxidation (LPO), protein tyrosine nitration (PTN), total antioxidant capacity (TAC) and lactate concentration were compared in corals under heat stress with nitrate enrichment and under heat stress only, suggesting that nitrate enrichment (3 μM, 14 days experiment in the laboratory) exacerbates the heat stress response of corals (Fernandes de Barros Marangoni et al., 2020). And nitrate enrichment (4 μM, 18 months in the field) causes doubles bleaching prevalence in *Acropora* corals under heat stress (Burkepile et al., 2020). Conversely, ammonium enrichment (2 μM, 20 days in the laboratory) has been demonstrated that is an important source of nitrogen to maintain photosynthesis and photosynthetic product translocation in Symbiodiniaceae to help alleviate the physiological response of the coral holobiont under heat stress (Ezzat et al., 2019; Fernandes de Barros Marangoni et al., 2020). From a transcriptome perspective, it was found that ammonium enrichment (10 μM, 12 h in the laboratory) can reduce the expression level of genes associated with apoptosis and cell death in corals under heat stress (Zhou et al., 2017).

Lipids are essential organic molecules in corals, accounting for ~30 % of their dry organic matter (Imbs, 2013). Fatty acids (FA) are the basic building blocks of lipids (Rocker et al., 2019). FA are often used as biomarkers for studies on coral health conditions (Rocker et al., 2019), symbiotic relationships between the coral host and its Symbiodiniaceae (Tchernov et al., 2004), coral bleaching and recovery (Sikorskaya and Imbs, 2020), impacts of environmental changes on corals (Kim et al., 2021), and coral reproduction (Grinyó et al., 2018). In a coral holobiont, lipid synthesis is mainly accomplished by Symbiodiniaceae (Patton et al., 1977). Saturated FA (SFA), the main storage lipids, can be efficiently metabolised for rapid energy production (Tocher, 2003). Polyunsaturated FA (PUFA) are major structural lipids, and the main lipids that regulate coral metabolism (Tocher, 2003). They are involved in physiological activities such as immune and inflammatory responses, neurological functions, reproduction, and maintenance of cell membrane fluidity (Tocher, 2003; Rocker et al., 2019). Symbiodiniaceae as algae possess Δ12 and Δ15 desaturases, which are lacking in coral hosts. 18:2n-6 (linoleic acid; LA) and 18:3n-3 (linolenic acid; ALA), as FA precursors for n-6 and n-3 PUFA, are synthesised in Symbiodiniaceae because Δ12 and Δ15 desaturases are required for the synthesis process (Bachok et al., 2006; Imbs, 2013). Healthy symbiotic coral hosts are rich in 18:3n-6 (gamma-linolenic acid; GLA) and 18:4n-3 (stearidonic acid; SDA), compared to the contents <1 % in bleached corals and ahermatypic corals (Bachok et al., 2006; Imbs et al., 2014). Thus, GLA and SDA are considered biomarkers of the presence of Symbiodiniaceae in the coral host (Imbs, 2013). Similarly, 20:4n-6 (arachidonic acid; ARA) in Symbiodiniaceae is considered a biomarker for symbiosis between Symbiodiniaceae and the coral host (Chen et al., 2015), because ARA is not detected in isolated cultures of Symbiodiniaceae nor in free-living marine dinoflagellates (Suzuki et al., 2003; Imbs et al., 2014). ARA is also an FA precursor necessary for pro-inflammatory hormone synthesis, and it is responsible for water transport across membranes (Tocher, 2003). 20:5n-3 (eicosapentaenoic acid; EPA) plays a role in anti-inflammatory processes, and 22:6n-3 (docosahexaenoic acid; DHA) is beneficial for the fluidity of cell membranes (Rocker et al., 2019). These molecules are essential n-3 and n-6 PUFA that drive inflammatory responses (Tocher et al., 2019; Kim et al., 2021).

However, the effects of nitrate and ammonium enrichment on FA biosynthesis pathways and translocation in coral hosts and Symbiodiniaceae under heat stress have not yet been studied. To fill this gap, we applied gas chromatography mass spectrometry (GC–MS) to explore differences in the effects of nitrate and ammonium enrichment on FA composition in *Acropora hyacinthus* coral host and Symbiodiniaceae under heat stress. *A. hyacinthus* is a common competitive coral (Darling et al., 2012). We optimised methods for isolation of Symbiodiniaceae from the coral host, and separate FA extraction from the coral host and Symbiodiniaceae. Specific FA ratios (n-3:n-6 PUFA, EPA:ARA and DHA:ARA) and certain FA biomarkers (SFA, PUFA, GLA, SDA and ARA) were used to determine the degree of the stress responses, health status, and stability of the symbiotic relationship between the coral host and Symbiodiniaceae.

This study aimed to identify (a) whether nitrate and ammonium enrichment under heat stress caused differential changes in FA associated with the stress responses, symbiotic relationships, and health status in coral hosts and Symbiodiniaceae, and (b) whether nitrate and ammonium enrichment under heat stress had different effects on FA biosynthesis and translocation in coral hosts and Symbiodiniaceae.

2. Materials & methods

2.1. Coral samples and experimental conditions

A. hyacinthus was collected from Wanning, Hainan Island, China. One *A. hyacinthus* coral was used and divided into small branches to avoid potential individual differences, and each sample contained approximately 4 to 5 small branches. The length of each coral branch is about 1.5 to 2 cm. All samples were restored in a 60-L aquarium for one week, with simulated natural light (12 h light, 12 h dark per day), and fed twice a week with *Artemia salina*. Before the start of the experiment, each sample was placed in a 5-L experimental beaker for 2 days to acclimatize. The same light and feeding were maintained throughout the experiment as during the recovery period.

The previous study has found the coverage of living reef corals is lowest when the DIN enrichment in seawater exceeded 9 μM (Duprey et al., 2016). Therefore, we used 9 μM as the experimental concentration for nitrate and ammonium enrichment. Six groups were established: control group (25 °C), nitrate enrichment group (25 °C and 9 μM nitrate enrichment), ammonium enrichment group (25 °C and 9 μM ammonium enrichment), heat stress group (30 °C), heat stress and nitrate enrichment group (30 °C and 9 μM nitrate enrichment), heat stress and ammonium enrichment group (30 °C and 9 μM ammonium enrichment). The same number of samples were placed in each group, and samples ($n = 4$) were taken on days 2 and 14 of the experiment to determine the Symbiodiniaceae density and the concentrations of

FA. To maintain experimental temperatures, heating rods (Yee 25 w) were used. And automatic titration pumps (Kamoer X1 PRO) were replenished with nitrate or ammonium to the water every 2 h to maintain the nitrogen concentrations stable. Artificial seawater was changed daily, and water quality was monitored.

2.2. Separation of the coral host and Symbiodiniaceae

We improved the separation method described in previous studies (Andersson et al., 2019; Kim et al., 2021). Coral samples were exposed to liquid nitrogen for 15 min then placed in 50 ml centrifuge tubes and mixed with 5 mM EDTA and butylated hydroxytoluene (BHT) at a 1000:1 (*v*/v) ratio. Samples were homogenised on ice with a tissue grinder for 1 min and the skeleton was removed. The homogenate was centrifuged for 1 h at 4 °C at 925*g*. A drop of supernatant was observed under a microscope to confirm that no Symbiodiniaceae cells remained in the coral host tissue sample. The supernatant was then transferred to another 50 ml tube and kept in the freezer at −80 °C. The pellet was resuspended in ultrapure water and centrifuged at 10*g* for 30 s to remove carbonate residue. A 10 μl sample of the supernatant was used to calculate the Symbiodiniaceae density (Section 2.4). The supernatant was also then transferred to another 50 ml tube and kept in the freezer at −80 °C. Before lipid extraction, samples were freeze-dried and kept as powder. The dry weight of the sample powder was measured.

2.3. Extraction and determination of lipid and FA

We improved the extraction method described in previous studies (Folch et al., 1957; Bachok et al., 2006; Revel et al., 2016; Kim et al., 2021). Sample powder was homogenised with ultrapure water, followed by sequential addition of methanol, chloroform and 0.9 % NaCl (1:2:0.6, *v*/v/v). After mixing, samples were incubated overnight at 4 °C in the dark. The next day, samples were centrifuged at 650*g* for 5 min, the methanol supernatant was removed, and chloroform, methanol and 0.9 % NaCl (2:1:0.6, *v*/*v*/v) were added for washing. After mixing, the supernatant was removed by centrifugation at 650*g* for 5 min, and methanol and 0.9 % NaCl (1:1, v/v) were added for washing. The supernatant was removed by centrifugation at 650*g* for 5 min, and after two repetitions, the extract was transferred to a brown bottle and dried under nitrogen gas to obtain total lipid (TL), and weighed. The purpose of measuring total lipid content is to normalize and quantify FA concentrations.

TL samples were spiked with internal standard (19:0 methyl nonadecanoate, 0.25 μg/l in dichloromethane), chloroform and 0.5 M KOH/CH_3OH (1:2.5, *v*/v) were added and mixed, and samples were heated in a water bath at 80 °C for 2 h. After cooling, 10 % BF_3-CH_3OH was added, mixed, and heated in a water bath at 80 °C for 2 h to form FA methyl esters (FAMEs). When cooled, saturated NaCl was added and FAMEs were extracted three times with n-hexane.

Extracted FAMEs were dried completely under nitrogen gas, resuspended in dichloromethane, and analysed by gas chromatography mass spectrometry (GC–MS) under electron ionisation (EI) mode. GC–MS (using an Agilent 5977B mass selective detector interfaced with an Agilent 8860 GC instrument) was performed using a DB-225 ms capillary column (30 m × 0.25 mm × 0.25 μm; Agilent, USA) to detect FAMEs. Helium (He) was used as the carrier gas with a flow rate of 1 ml min^{-1}. A 1 μl sample of derivatised sample was injected into the GC–MS. The column temperature was initially held at 80 °C, then increased to 240 °C at a temperature gradient of 3 °C min^{-1}, and held for 4 min. An external standard mix (37-component FAME mix; Shanghai Yuanye, China) and the internal standard were used to identify and quantify each FAME. For quantification, the dry weight of TL was used to normalize the concentrations of FAMEs (μg mg^{-1}).

2.4. Calculation of Symbiodiniaceae density

We performed analyses of Symbiodiniaceae density following standard procedures (Li et al., 2008). First, a correlation standard curve was established between the weight and surface area of aluminium foil. Then, the surface of coral samples was wrapped by aluminium foil. The surface area of the coral sample was calculated using the standard curve. From Symbiodiniaceae supernatant obtained in Section 2.2, 10 μl of Symbiodiniaceae supernatant was pipetted and diluted to 100 μl with ultrapure water. Symbiodiniaceae cell counts were measured by replicate hemacytometer counting ($n = 10$) under a microscope. Finally, calculation of Symbiodiniaceae density ($\times 10^6$ cells cm^{-2}) was performed.

2.5. Data analysis

OriginPro (version 2022b, USA) was used to perform data analysis and statistics. One-way analysis of variance (ANOVA) and Tukey's post-hoc test were used to determine statistical differences in FA concentrations, FA ratios, and Symbiodiniaceae density between treatment groups. Differences were considered significant at $p < 0.05$.

3. Results & discussion

3.1. Changes in Symbiodiniaceae density and coral bleaching

After 2 days of stress, Symbiodiniaceae density was not significantly different between control and treatment groups ($p > 0.05$; Fig. 1). After 14 days, heat stress caused a significant reduction ($p < 0.05$) in Symbiodiniaceae density, and nitrate enrichment exacerbated the reduction ($p < 0.05$). Nitrate enrichment under heat stress resulted in the lowest Symbiodiniaceae density. However, Symbiodiniaceae density was increased significantly in the ammonium enrichment group ($p < 0.05$), and this density was the highest among all groups. Also, ammonium enrichment under heat stress was beneficial for *A. hyacinthus* to maintain Symbiodiniaceae density ($p < 0.05$, compared with the heat stress group).

Changes in the overall state of coral during the experiment were documented through photographs (Supplementary Fig. S1). Coral samples from the heat stress group and the heat and nitrate enrichment co-stress group showed a clear tendency toward bleaching after 14 days of treatment. These changes corresponded to the trend in Symbiodiniaceae density.

3.2. Changes in FA of the coral host and Symbiodiniaceae under different treatment groups

3.2.1. Total lipid

On the 2nd day of the experiment, TL concentrations of coral hosts were increased significantly ($p < 0.05$) in heat stress and ammonium enrichment groups (Fig. 2). Conversely, TL concentrations of Symbiodiniaceae were significantly reduced ($p < 0.05$) in all nitrogen enrichment groups. This phenomenon might be due to short-term heat stress and ammonium enrichment promoted translocation rate of photosynthetic products in Symbiodiniaceae (Rosic et al., 2014; Ezzat et al., 2015; Rädecker et al., 2021), subsequently might render Symbiodiniaceae to transfer more lipids to coral hosts (Dunn et al., 2012). Therefore, heat and ammonium enrichment co-stress groups had the highest TL concentrations for the coral host (Fig. 2). Conversely, short-term nitrate enrichment led to a significant reduction ($p < 0.05$) in TL content in the coral host and Symbiodiniaceae. This is likely because the conversion of nitrate to ammonium by Symbiodiniaceae requires energy (Dagenais-Bellefeuille and Morse, 2013).

However, constant heat stress significantly reduced ($p < 0.05$) the energy substances of coral hosts and Symbiodiniaceae (Fig. 2). Nitrate enrichment after 14 days of heat stress further exacerbated ($p < 0.05$) the energy costs of both Symbiodiniaceae and coral host (Fig. 2). Ammonium is an essential N substrate required for the resilience of corals to heat stress, thus ammonium enrichment can alleviate the energy requirements of Symbiodiniaceae under heat stress (Ezzat et al., 2019), which might be the reason that no significant change in Symbiodiniaceae density between the ammonium enrichment and heat co-stress group and the control group (Fig. 1). However, the energy consumption required to maintain normal survival induced Symbiodiniaceae to optionally reduce the transfer of

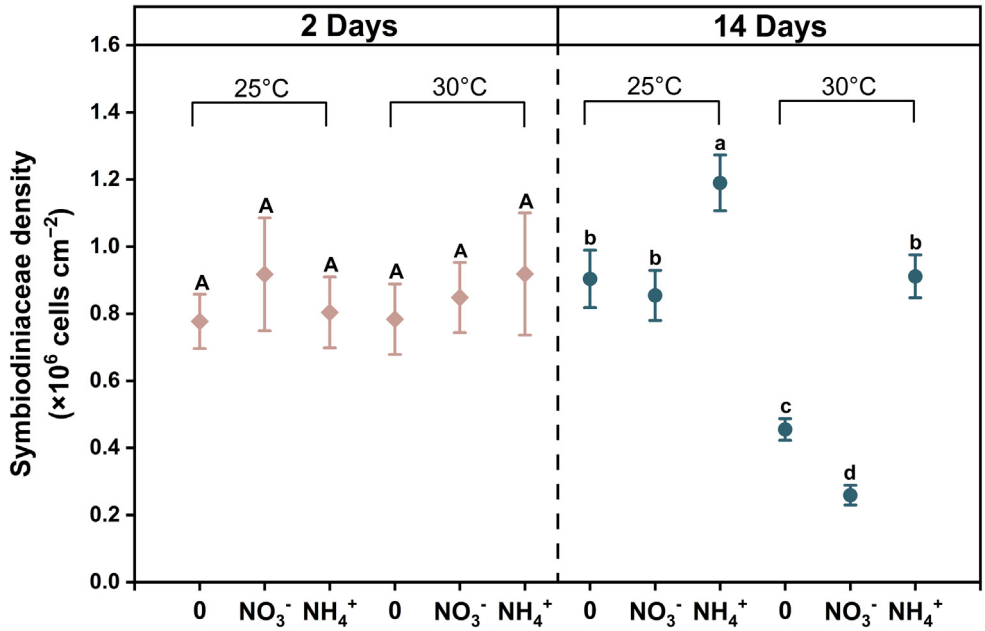

Fig. 1. Changes in Symbiodiniaceae density activity based on each treatment group over 2 and 14 days. All data points are mean ± SE (n = 10). 0 indicates no nitrogen addition; NO_3^- indicates 9 μM nitrate addition; NH_4^+ indicates 9 μM ammonium addition. Capital case and lower case letters indicate significant differences between treatment groups for the different days, respectively (One-Way ANOVA, Turkey's test, $p < 0.05$).

lipids as energy material to coral hosts (Morris et al., 2019). Therefore, TL contents of the coral host and Symbiodiniaceae were significantly reduced (Fig. 2). Nevertheless, TL could not completely explain the differential effects of nitrate and ammonium enrichment on coral hosts and Symbiodiniaceae under heat stress, hence we explored FA indicators.

3.2.2. FA involved in stress responses

The changes in SFA and PUFA of the 14th of experiment were more obvious compared to those of the 2nd day, indicating that the longer treatment time had a greater effect on SFA and PUFA. On the 14th day, SFA and PUFA levels in coral hosts were significantly reduced (compared to the control group, $p < 0.05$) in all treatment groups except for the nitrate enrichment group (Fig. 3). This pattern of change was consistent with the 2nd day experiment. SFA and PUFA are involved in maintenance of cell membrane fluidity (Tocher, 2003). Heat and nitrate enrichment co-stressed coral hosts showed a significant increase (compared to heat stress group, $p < 0.05$) in SFA and PUFA content, potentially due to cell membranes might become more rigid and viscous to slow the entry of excessive nitrate (Liu et al., 2020). Notably, SFA and PUFA concentrations of the coral host were increased in the control group after 14 days compared to those after 2 days, which might be due to the feeding during the experiment (Kim et al., 2021).

SFA are usually consumed for rapid energy production, while PUFA are involved in immune and inflammatory responses (Tocher, 2003). PUFA also are the main target of ROS for cell damage (Lesser, 2006). The decline in PUFA content under heat stress (Fig. 3B) might indicate the accumulation of ROS in both coral host and Symbiodiniaceae cells (Downs et al., 2013).

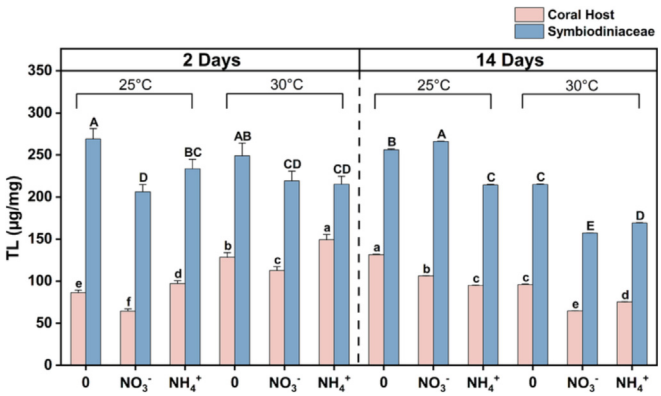

Fig. 2. Changes in total lipid based on each treatment group at 2 and 14 days. All data points are mean ± SD (n = 4). Capital case and lower case letters indicate significant differences between treatment groups for coral host and Symbiodiniaceae, respectively (One-Way ANOVA, Turkey's test, $p < 0.05$).

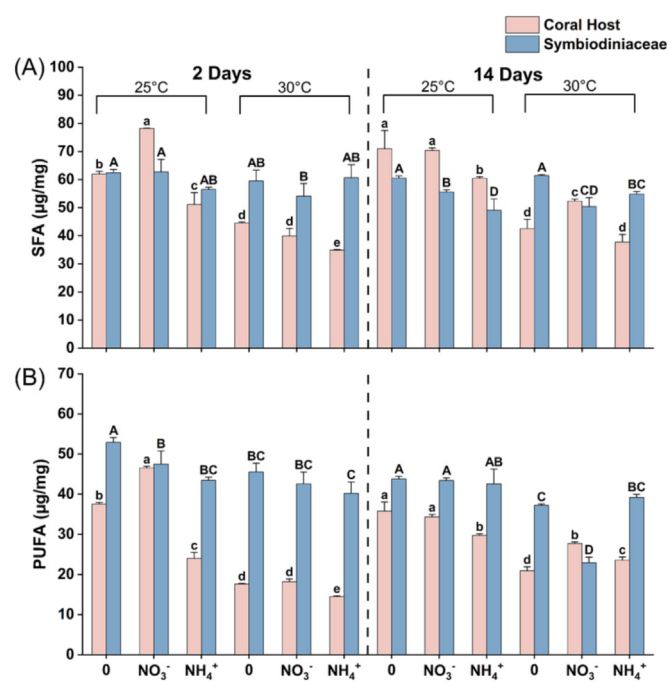

Fig. 3. Saturated fatty acids (A) and polyunsaturated (B) fatty acids of the coral host and Symbiodiniaceae at 2 and 14 days under different treatment groups. All data points are mean ± SD (n = 4). Capital letters and lower letters indicate significant differences between treatment groups for host and Symbiodiniaceae respectively (One-Way ANOVA, Turkey's test, $p < 0.05$).

Heat stress causes damage to the mitochondria of coral hosts, and the chloroplasts and mitochondria of Symbiodiniaceae (Weis, 2008). Meanwhile, the metabolism of coral hosts was accelerated (Brown et al., 2004) and the production and translocation of photosynthetic products were reduced by Symbiodiniaceae under heat stress (Wooldridge, 2014). Thus, coral hosts were forced to catabolize SFA (Fig. 3A) to maintain their physiological activities and the stability of the intracellular environment (Hillyer et al., 2017).

Nitrate enrichment under heat stress not only fails to alleviate thermal damage of Symbiodiniaceae, but also discourages its energy intake and further increases reactive nitrogen species (RNS) (Dagenais-Bellefeuille and Morse, 2013; Ezzat et al., 2015). Therefore, nitrate enrichment exacerbated the heat stress response of Symbiodiniaceae, and induced the lowest PUFA levels in Symbiodiniaceae (Fig. 3B). This resulted in the expulsion of the largest number of Symbiodiniaceae (Fig. 1) and a reduction in TL levels in coral hosts (Fig. 2). Excessive expulsion of Symbiodiniaceae also led to a significant reduction in SFA in Symbiodiniaceae (Fig. 3A) and the accumulation of n-6 PUFA in the coral host (Table S3). The inflammatory response can be induced by accumulation of n-6 PUFA (Tocher, 2003; Russo, 2009), hence nitrate enrichment under heat stress exacerbated the inflammatory response in the coral host. Meanwhile, coral hosts exposed to a low Symbiodiniaceae density for long periods of time will die due to energy starvation (Imbs and Yakovleva, 2012).

Conversely, ammonium enrichment has been demonstrated that is an important source of nitrogen to maintain photosynthesis and photosynthetic product translocation in Symbiodiniaceae to help alleviate the physiological response of the coral holobiont under heat stress (Ezzat et al., 2019; Fernandes de Barros Marangoni et al., 2020). Therefore, ammonium enrichment contributed to alleviation of the heat stress response in the coral host and Symbiodiniaceae (Zhou et al., 2017). Thus, ammonium enrichment led to higher PUFA levels in coral hosts and Symbiodiniaceae under heat stress (Fig. 3B). Ammonium enrichment also drove a further acceleration in the metabolism of the coral host under heat stress (Rosic et al., 2014; Fernandes de Barros Marangoni et al., 2020), and SFA were rapidly consumed for energy production (Fig. 3A). Meanwhile, ammonium enrichment was beneficial to the recovery and growth of Symbiodiniaceae under heat stress (Fig. 1), hence SFA were consumed by Symbiodiniaceae to repair cell damage and support growth (Fernandes de Barros Marangoni et al., 2020).

In summary, we found that the most significant reductions in SFA and PUFA of coral hosts and Symbiodiniaceae under heat stress. The reduction of PUFA in Symbiodiniaceae were exacerbated by nitrate enrichment, while ammonium enrichment alleviated the effects of heat stress on PUFA in coral hosts and Symbiodiniaceae. Since reduction in PUFA was associated with the accumulation of ROS (Downs et al., 2013), nitrate enrichment exacerbated the heat stress response in Symbiodiniaceae and ammonium enrichment alleviated the heat stress response in both coral hosts and Symbiodiniaceae.

3.2.3. FA associated with symbiotic relationships

The changes in 18:3n-6 and 18:4n-3 were more pronounced with increasing treatment time and the trend on 2nd day were similar to those on 14th day. On 14th day, both nitrate and ammonium enrichment under heat stress intensified the reduction in 18:3n-6 and 18:4n-3 concentrations of coral hosts (compared to the heat stress group, $p < 0.05$; Fig. 4A, B). Nitrate enrichment group and heat stress group showed significantly increased (compared to the control group, $p < 0.05$) 18:3n-6 concentrations of Symbiodiniaceae (Fig. 4A). By contrast, 18:3n-6 concentration of Symbiodiniaceae was significantly reduced (compared to the heat stress group, $p < 0.05$) in nitrogen enrichment group under heat stress (Fig. 4A). The lowest concentrations of 18:3n-6 and 18:4n-3 of Symbiodiniaceae were both observed in the heat and nitrate co-stress group (Fig. 4A, B), while 18:4n-3 concentration of the other treatment groups was significantly increased ($p < 0.05$). Both 18:3n-6 and 18:4n-3 are produced and translocated to coral hosts by Symbiodiniaceae (Imbs, 2013; Sikorskaya and Imbs, 2020), and they can be synthesised by coral hosts themselves,

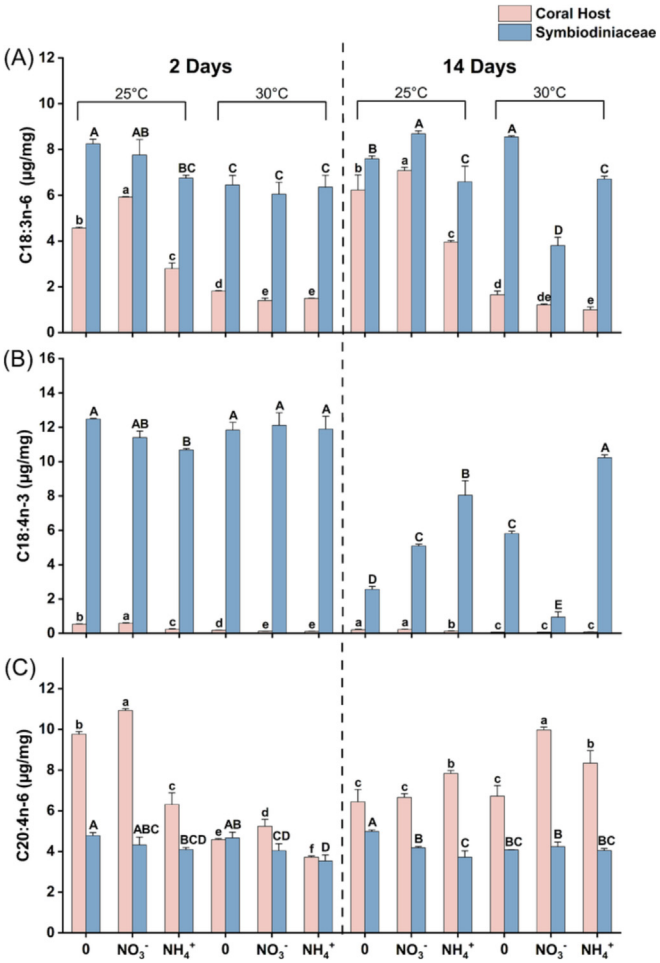

Fig. 4. Effects on the symbiotic relationship between the coral host and Symbiodiniaceae at 2 and 14 days under different treatment groups. Fatty acids 18:3n-3 (A), 18:4n-3 (B), 20:4n-6 (C) as relevant indicators. All data points are mean ± SD (n = 4). Capital case and lower case letters indicate significant differences between treatment groups for host and Symbiodiniaceae respectively (One-Way ANOVA, Turkey's test, $p < 0.05$).

but with very low efficiency (Imbs et al., 2010). Thus, heat stress might inhibit the translocation of 18:3n-6 and 18:4n-3 from Symbiodiniaceae to coral hosts, and the inhibition was exacerbated by nitrate enrichment.

However, changes in 20:4n-6 of coral hosts in each treatment group were inconsistent during the experiment. On 2nd day, ammonium enrichment under heat stress inhibited the synthesis of 20:4n-6 in coral hosts (compared to the heat stress group, $p < 0.05$), while nitrate enrichment under heat stress promoted the synthesis of 20:4n-6 (compared to the heat stress group, $p < 0.05$). On 14th day, nitrogen enrichment promoted the significant accumulation of 20:4n-6 in coral hosts (compared to the heat stress group, $p < 0.05$; Fig. 4C). 20:4n-6 mainly synthesised from the metabolism of 18:3n-6 by the coral host (Imbs et al., 2010). Thus, the early stages of experiment, Symbiodiniaceae might have reduced the amount of 18:3n-6 provided to coral hosts (Fig. 4A), resulting in the inhibition of 20:4n-6 synthesis in coral hosts. By contrast, the change in 20:4n-6 of Symbiodiniaceae was more visible on 14th day. On 2nd day, only nitrogen enrichment under heat stress significantly reduced 20:4n-6 concentration of Symbiodiniaceae (compared to the control group, $p < 0.05$; Fig. 4C), while 20:4n-6 was significantly reduced in all treatment groups at day 14 (compared to the control group, $p < 0.05$; Fig. 4C). 20:4n-6 acts as a biomarker for symbiosis between coral hosts and Symbiodiniaceae (Chen et al., 2015), so Fig. 4C shows that all treatment groups had an effect (compare to the control group) on the translocation of 20:4n-6 from the coral host to Symbiodiniaceae.

Heat stress caused damage to the membrane lipids of chloroplasts of Symbiodiniaceae (Tchernov et al., 2004). 18:3n-6 as a fatty acid precursor for the synthesis of pro-inflammatory eicosanoids (Tocher, 2003; Russo, 2009), inflammatory response led to a significant increase in 18:3n-6 concentration of Symbiodiniaceae (Fig. 4A). The immune responses of Symbiodiniaceae drive the retention of 18:3n-6 and 18:4n-3 that would normally be translocated to the coral host (Fernandes de Barros Marangoni et al., 2020). Thus, heat stress could have a significant negative impact on the translocation of 18:3n-6 and 18:4n-3 biosynthesised by Symbiodiniaceae to the coral host. This negative effect was already evident in the early stages of the experiment (Fig. 4A, B). In fact, the symbiotic relationship between the coral host and Symbiodiniaceae is dependent on the translocation of energetic substances (especially lipids) (Petrou et al., 2021), hence this negative effect would disrupt this symbiotic relationship.

Also, nitrate enrichment appeared to exacerbate disruption of the symbiotic relationship by heat stress (Fig. 4A, B). Nitrate enrichment could make it more difficult for Symbiodiniaceae to synthesise 18:4n-3 (Fig. 4B), which would be highly detrimental for Symbiodiniaceae under heat stress (Dagenais-Bellefeuille and Morse, 2013; Ezzat et al., 2015). This would make it difficult for Symbiodiniaceae to survive following heat stress responses. Meanwhile, heat stress could promote the rapid assimilation of nitrate by Symbiodiniaceae and cause it to reduce the amount of organic derivatives translocated to the coral host (Baker et al., 2018). This would shift the symbiosis between the coral host and Symbiodiniaceae to parasitism. As a result, there was a significant loss of Symbiodiniaceae (Fig. 1).

By contrast, ammonium enrichment is beneficial for Symbiodiniaceae under heat stress. It has been studied that in corals, coral hosts can use ammonium (NH_4^+) to produce NH_3 and translocate it to Symbiodiniaceae, since the intracellular environment of Symbiodiniaceae is acidic and can convert NH_3 to ammonium (NH_4^+) for use (Thies et al., 2022). Thus, ammonium enrichment provides energy that Symbiodiniaceae can use to repair themselves and grow, and FA synthesis is restored (Fernandes de Barros Marangoni et al., 2020). Symbiodiniaceae also transferred some of the newly synthesised FA to the coral host (Table S3), resulting in some relief of the heat stress responses (Section 3.2.2, Fig. 3B). This also suggests that ammonium enrichment might contribute to repairing the symbiotic relationship between the coral host and Symbiodiniaceae. Notably, the reduced 18:4n-3 content of Symbiodiniaceae in the 14-day control group might be due to DHA synthesis and translocation to the coral host (Table S3).

3.2.4. FA ratios indicate health conditions

On 2nd day of the experiment, the ratios of n-3:n-6 PUFA, EPA:ARA (20:5n-3 to 20:4n-6) and DHA:ARA (22:6n-3 to 20:4n-6) of coral hosts and Symbiodiniaceae were significantly reduced ($p < 0.05$) in all treatment groups (Fig. 5) except the nitrate enrichment group (Fig. 5C). The lowest ratios were observed in the heat and nitrate enrichment co-stress group. The trends in all ratios of coral hosts remained at day 14 of the experiment. However, the trends in all ratios of Symbiodiniaceae on 14th day were not consistent with those on 2nd day. Ammonium enrichment significantly increased ($p < 0.05$) all ratios, regardless of heat stress (Fig. 5). Heat stress significantly reduced ($p < 0.05$) all ratios (Fig. 5), and nitrate enrichment significantly exacerbated ($p < 0.05$) this reduction, while ammonium enrichment significantly mitigated ($p < 0.05$) the reduction. All ratios in Symbiodiniaceae were highest in the ammonium enrichment group, and the all lowest ratios were observed in the heat and nitrate co-stress group.

Herein, we utilised n-3:n-6 PUFA, EPA:ARA and DHA:ARA ratios to assess the health condition of the coral host and Symbiodiniaceae. The three ratios were used simultaneously to remove chance factors, and the trends in all three ratios were found to be relatively consistent. n-3 PUFA and n-6 PUFA are closely related to inflammatory responses (Tocher et al., 2019; Kim et al., 2021); n-3 PUFA is associated with the anti-inflammatory response and n-6 PUFA is associated with the pro-inflammatory response (Tocher, 2003; Russo, 2009). ARA is the n-6 PUFA precursor required for a pro-inflammatory response to occur (Tocher, 2003; Imbs et al., 2010).

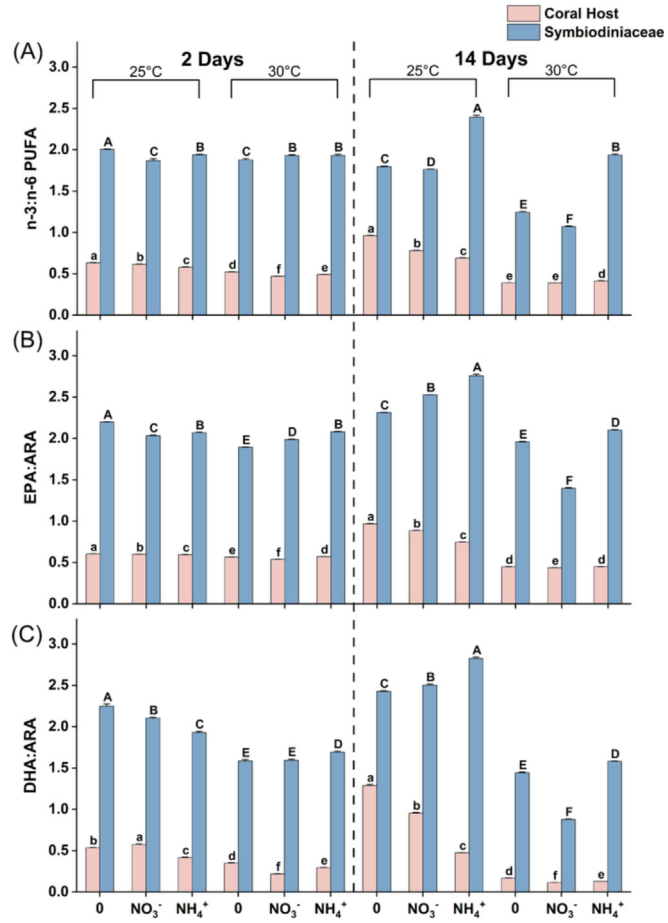

Fig. 5. Fatty acid ratios of n-3:n-6, EPA:ARA, DHA:ARA as indicators of the health status of the coral host and the Symbiodiniaceae at 2 and 14 days under different treatment groups. All data points are mean ± SD (n = 4). Capital case and lower case letters indicate significant differences between treatment groups for host and Symbiodiniaceae respectively (One-Way ANOVA, Turkey's test, $p < 0.05$).

EPA and DHA are n-3 PUFA associated with anti-inflammatory responses (Tocher, 2003). DHA has a positive effect on cellular processes such as enzyme activity, immune functions and adaptation to heat stress (Bergé and Barnathan, 2005; Mies et al., 2017). Thus, higher values of ratios indicate a better health compared with lower values.

All treatment groups resulted in a worse health status for the coral host, and this was evident in the early stages of the experiment (Fig. 5). Heat stress significantly inhibited synthesis of n-3 PUFA in Symbiodiniaceae, led to a reduction in n-3 PUFA content of coral hosts (Table S3). Thus, heat stress lowered the health of both coral hosts and Symbiodiniaceae (Hillyer et al., 2017; Petrou et al., 2021). Nitrate enrichment made the heat damage even more severe (Fig. 5). Nitrate enrichment further inhibited n-3 PUFA synthesis of Symbiodiniaceae under heat stress, and promoted the accumulation of n-6 PUFA in coral host (Table S3). This leading to a worse health status of Symbiodiniaceae and coral hosts (Fig. 5). However, ammonium enrichment benefited the biosynthesis of n-3 PUFA in the coral host, and greatly reversed the damage caused by heat stress to Symbiodiniaceae by promoting FA synthesis (LeKieffre et al., 2020) (Table S3), thereby promoting the recovery of health in both coral hosts and Symbiodiniaceae (Fig. 5).

3.2.5. FA biosynthesis and translocation

In a coral holobiont, 18:3n-3 and 18:2n-6 as synthetic precursors of n-3 and n-6 PUFA series are synthesised by Symbiodiniaceae because only Symbiodiniaceae possesses the Δ12 and Δ15 desaturases (Bachok et al., 2006; Imbs, 2013). Also, bleached corals and ahermatypic corals had

extremely low levels of 18:3n-6 and 18:4n-3 compared to healthy symbiotic corals, suggesting that 18:3n-6 and 18:4n-3 were supplied to coral hosts by Symbiodiniaceae (Bachok et al., 2006; Imbs et al., 2014). 20:4n-6 and 22:4n-6 were mainly biosynthesised by coral hosts and translocated to Symbiodiniaceae since coral hosts (20:4n-6, 6.44 ± 0.61 μg mg^{-1}; 22:4n-6, 4.32 ± 0.25 μg mg^{-1}) contain more of them than Symbiodiniaceae (20:4n-6, 4.99 ± 0.07 μg mg^{-1}; 22:4n-6, 1.98 ± 0.03 μg mg^{-1}). In addition, 20:4n-6 is not detected in isolated cultures of Symbiodiniaceae nor in free-living marine dinoflagellates (Suzuki et al., 2003; Imbs et al., 2014). 20:5n-3 was mainly biosynthesised by Symbiodiniaceae and translocated to coral hosts, owing to the fact that they are synthesised in more quantities in Symbiodiniaceae (20:5n-3, 11.53 ± 0.15 μg mg^{-1}) than in coral hosts (20:5n-3, 6.22 ± 0.56 μg mg^{-1}). Meanwhile, 22:5n-3 and 22:6n-3 are not only biosynthesized and supplied to coral hosts (22:5n-3, 2.81 ± 0.18 μg mg^{-1}; 22:6n-3, 8.28 ± 0.69 μg mg^{-1}) by Symbiodiniaceae (22:5n-3, 1.87 ± 0.03 μg mg^{-1}; 22:6n-3, 12.10 ± 0.21 μg mg^{-1}), but also obtained by ingestion of coral hosts themselves (Kim et al., 2021). Notably, the lower 22:6n-3 concentration of coral hosts than that of Symbiodiniaceae is due to limited biosynthesis of 22:6n-3 in coral hosts (Revel et al., 2016). As shown in Fig. 6, we combined previous studies with FA data of control group (Table S3) obtained in the present work to propose a possible biosynthesis pathway and translocation of FA for the *A. hyacinthus* coral host and Symbiodiniaceae (Imbs et al., 2010; Imbs, 2013; Revel et al., 2016). The n-3 and n-6 PUFA synthesis pathways are in competition because they use the same biosynthetic enzymes, and which pathway is dominant depends on the availability of FA precursors (Nakamura and Nara, 2004). Thus, changes in the FA biosynthesis pathways in Symbiodiniaceae can affect FA acid biosynthesis pathways in the coral host.

Under heat stress (Fig. 6A), both the translocation and biosynthesis of major FA were significantly reduced ($p < 0.05$) in the coral host compared with the control group. In particular, accumulation of 22:4n-6 in the coral host and acquisition of 18:3n-3 by the coral host was significantly increased ($p < 0.05$). n-6 PUFA biosynthesis was significantly increased in the coral host, whereas n-3 PUFA biosynthesis was significantly reduced. This suggests that heat stress leads to a predominance of the n-6 PUFA biosynthesis pathway in coral hosts. By contrast, Symbiodiniaceae retained C18 FA under these conditions. There was significant accumulation ($p < 0.05$) of 18:2n-6, 18:3n-3, 18:3n-6 and 18:4n-3 in Symbiodiniaceae. This finding is consistent with Symbiodiniaceae storing energy substances rather than undergoing translocation under heat stress (Fig. 3A), as reported in previous studies (Hillyer et al., 2018). This probably reflects damage caused

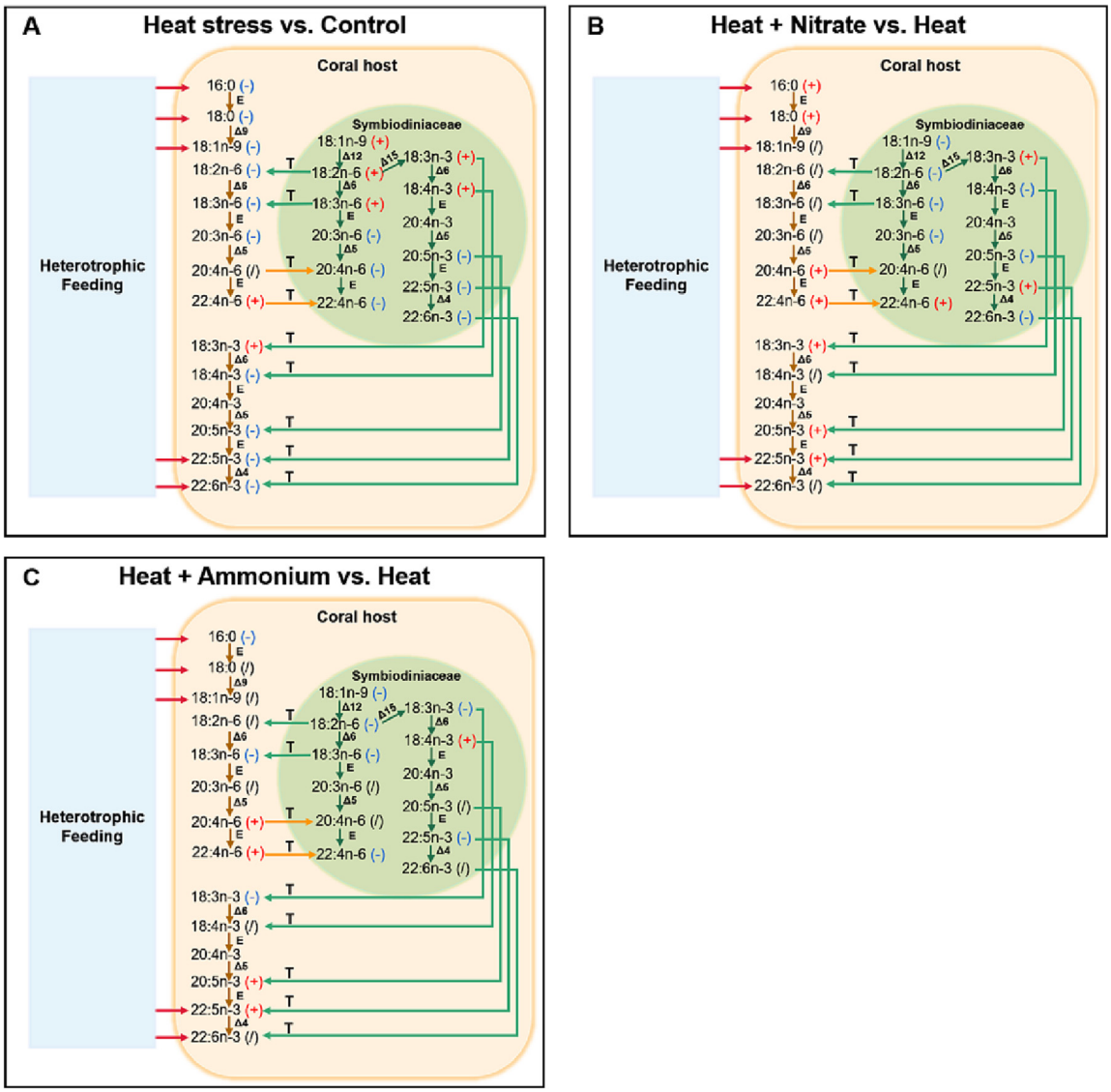

Fig. 6. The fatty acids biosynthesis pathway and potential translocation options in *A. hyacinthus* coral host and Symbiodiniaceae. (−) indicates a significant reduction in concentration, (+) indicates a significant increase in concentration, (/) indicates no significant change; E indicates elongase, T indicates translocate; Δ4, Δ5, Δ6, Δ9, Δ12 and Δ15 are desaturase (n = 4, One-Way ANOVA, Turkey's test, $p < 0.05$).

by heat stress to the desaturation and/or elongation process of FA in Symbiodiniaceae (Díaz-Almeyda et al., 2011). Stagnation of the FA biosynthesis pathway in Symbiodiniaceae affects the n-3 PUFA biosynthesis pathway in the coral host. Also, heat stress can inhibit the feeding capacity of the *A. hyacinthus* coral host (Traylor-Knowles et al., 2017). These effects combine to make it very difficult for the coral host to obtain energy to fight heat stress responses.

Under nitrate-enriched environments, n-3 PUFA biosynthesis was inhibited in Symbiodiniaceae (Table S3). This could make it more difficult for coral hosts to obtain n-3 PUFA, which in turn can induce inflammatory responses in coral hosts, and ultimately increase the sensitivity of corals to heat stress (Wiedenmann et al., 2013; Vega Thurber et al., 2014; Donovan et al., 2020). Under heat and nitrate enrichment co-stress, levels of 16:0, 18:0, 20:4n-6, 22:4n-6, 18:3n-3, 20:5n-3 and 22:5n-3 were significantly increased ($p < 0.05$) in the coral host compared to the heat stress group (Fig. 6B). The n-6 PUFA biosynthesis pathway remained dominant in coral hosts in the heat and nitrate enrichment co-stress group, and accumulation of 22:4n-6 was significantly increased compared with the heat-stressed group (Fig. 6B). Additionally, the coral host may translocate accumulated n-6 PUFA to Symbiodiniaceae (Chen et al., 2015). Significant accumulation ($p < 0.05$) of 22:4n-6, 18:3n-3 and 22:5n-3 was observed in Symbiodiniaceae compared to the heat stress group. The significant increase in 22:4n-6 of Symbiodiniaceae may have exacerbated the inflammatory response of Symbiodiniaceae (Tocher, 2003; Russo, 2009). However, because the increased 22:4n-6 was not biosynthesised by Symbiodiniaceae, n-3 PUFA biosynthesis was significantly increased in Symbiodiniaceae to counteract the exacerbated inflammatory response (Tocher et al., 2019). This would also provide the coral host with 18:3n-3 to biosynthesise n-3 PUFA. However, due to the dominance of the n-6 PUFA biosynthesis pathway, n-3 PUFA biosynthesis in the coral host was minimal, and the increased 20:5n-3 and 22:5n-3 may have been obtained by translocation from Symbiodiniaceae. The small amount of n-3 PUFA acquired via translocation did not help the coral host to recover from heat stress (Fig. 5). Therefore, we speculate that nitrate enrichment under heat stress exacerbated the stress response of the coral host, and that n-6 PUFA were synthesised abundantly and translocated to Symbiodiniaceae. Furthermore, Symbiodiniaceae could not afford the additional n-6 PUFA due to heat and nitrate enrichment co-stress, and n-3 PUFA produced as a result of the immune response had to be translocated to the coral host, which exacerbated the stress response of Symbiodiniaceae. The small amount of n-3 PUFA obtained by the coral host was insufficient to counter the large amount of n-6 PUFA from biosynthesis, exacerbating the stress response of the coral host. Thus, nitrate enrichment under heat stress not only caused serious damage to the health status of both the coral host and Symbiodiniaceae (Fig. 5), but also led to a breakdown of their symbiotic relationship (Fig. 4).

Under ammonium-enriched environments, excess photosynthetic carbon produced by Symbiodiniaceae acts as a reserve for lipids in coral host cells (Dunn et al., 2012). Only 18:4n-3 was significantly accumulated ($p < 0.05$) in Symbiodiniaceae in the heat and ammonium enrichment group (Fig. 6C). Concentrations of 20:4n-6, 22:4n-6, 20:5n-3 and 22:5n-3 were significantly increased ($p < 0.05$) in the coral host in the heat and ammonium enrichment co-stressed group compared with the heat stress group. Therefore, ammonium enrichment under heat stress caused the n-3 PUFA biosynthesis pathway of Symbiodiniaceae to be dominant, and the synthesised n-3 PUFA were translocated to the coral host. The concentrations of 20:5n-3 and 22:5n-3 translocated from Symbiodiniaceae to the coral host were significantly increased compared with the heat stress group (Fig. 6C). Meanwhile, n-3 PUFA biosynthesis was increased significantly in the coral hosts. This suggests that the n-3 PUFA biosynthesis pathway in the coral host became operational, reversing the dominance of the n-6 PUFA biosynthesis pathway. This facilitated the recovery of the health status of coral hosts under heat stress (Fig. 5). Therefore, we conclude that ammonium enrichment under heat stress can promote the biosynthesis of n-3 PUFA, thereby contributing to the recovery of the coral host and the health of Symbiodiniaceae (Fig. 5). It also facilitated the translocation of n-3 PUFA from Symbiodiniaceae to the coral host, which was beneficial to the reconstitution of the symbiotic relationship (Fig. 4).

Based on the above findings, we propose that under stress, the coral host can obtain biosynthetic anti-inflammatory FA from Symbiodiniaceae by translocating pro-inflammatory FA to Symbiodiniaceae. However, if Symbiodiniaceae are already in an energy-deficient state, delivering additional anti-inflammatory FA to the coral host may lead to further deterioration of the health condition of Symbiodiniaceae. By contrast, if the FA biosynthesis pathway is gradually restored to normal, Symbiodiniaceae can synthesise plenty of anti-inflammatory FA to resist pro-inflammatory FA translocated from the coral host. Meanwhile, Symbiodiniaceae can translocate extra anti-inflammatory FA to the coral host to help restore health.

4. Conclusions

Seawater nitrogen enrichment caused by human activities has exacerbated the impact of global warming on reef-building corals. However, the role of different types of nitrogen enrichment on corals under heat stress is not yet clear. This study was the first to investigate the effects of nitrate and ammonium enrichment on *A. hyacinthus* coral hosts and their Symbiodiniaceae under heat stress from the perspective of FA biosynthesis pathways and translocation.

Our results revealed that heat stress promoted pro-inflammatory FA (n-6 PUFA) biosynthesis in coral hosts, and inhibited FA biosynthesis in Symbiodiniaceae. This could induce inflammatory responses and lead to deterioration in health status and symbiotic relationship between coral hosts and Symbiodiniaceae. Ammonium enrichment could promote anti-inflammatory FA (n-3 PUFA) biosynthesis in Symbiodiniaceae and reduce pro-inflammatory FA concentrations under heat stress, which was beneficial for the health recovery of Symbiodiniaceae and coral hosts under heat stress, and reconstruction of the symbiotic relationship. By contrast, nitrate enrichment under heat stress could further inhibit Symbiodiniaceae anti-inflammatory FA synthesis and promote pro-inflammatory FA synthesis in coral hosts. Therefore, nitrate enrichment aggravated the destruction of the symbiotic relationship between coral hosts and Symbiodiniaceae caused by heat stress, and had a more adverse effect on the health status of coral hosts and Symbiodiniaceae.

Our study confirmed that nitrate enrichment is toxic but ammonium is antidotal for reef-building corals under heat stress conditions, from the perspective of FA biosynthesis and translocation. Moreover, we proposed that nitrate enrichment may increase coral sensitivity to heat stress by inducing the inflammatory response of coral hosts through reduction of anti-inflammatory FA biosynthesis in Symbiodiniaceae. This provides new insights into the impact of human activities on coral reef ecosystems. In future research, we expect that lipidomics combined with other omics approaches will provide further insight into the mechanisms of coral bleaching. Additionally, research on FA in stress-tolerant corals under different environmental stress conditions is urgently needed.

CRediT authorship contribution statement

Jingjing Zhang: Conceptualization, Methodology, Validation, Visualization, Data curation, Formal analysis, Writing – original draft, Writing – review & editing. **Zanhui Huang:** Conceptualization, Validation, Project administration, Investigation, Writing – review & editing. **Yuanchao Li:** Methodology, Validation, Resources, Investigation, Writing – review & editing. **Dinghui Fu:** Investigation, Methodology, Visualization. **Qipei Li:** Methodology, Visualization, Writing – review & editing. **Lixin Pei:** Investigation, Software. **Yanwei Song:** Investigation, Software. **Liang Chen:** Investigation, Software. **Hongwei Zhao:** Conceptualization, Funding acquisition, Investigation, Supervision, Project administration, Writing – review & editing. **Shuh-Ji Kao:** Conceptualization, Writing – review & editing.

Data availability

Data will be made available on request.

Declaration of competing interest

The authors declare no conflict of interest in this article.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.scitotenv.2023.162834.

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